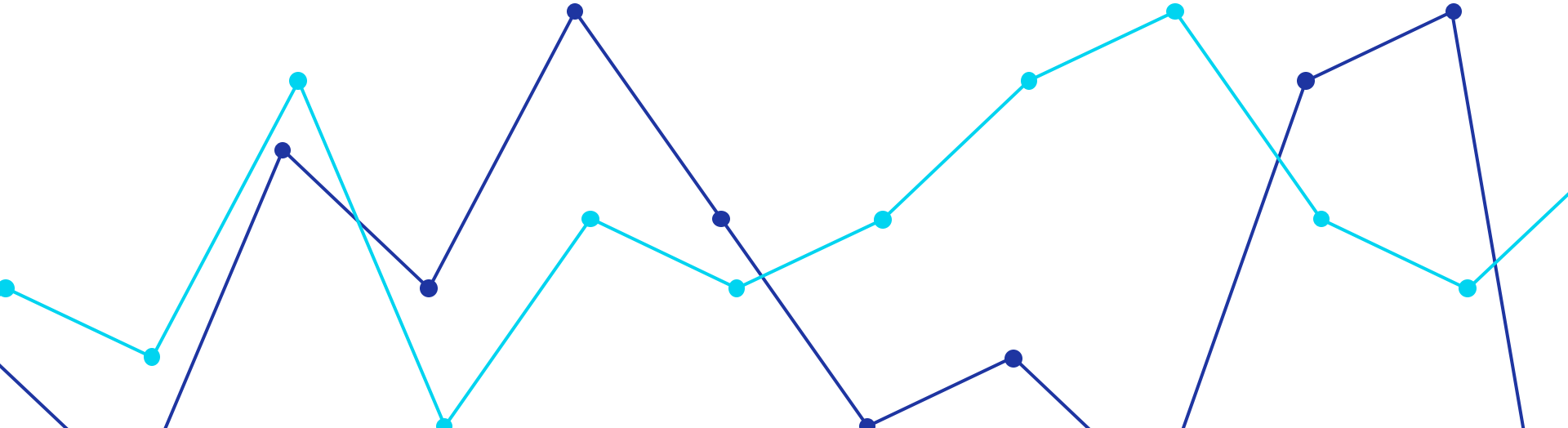


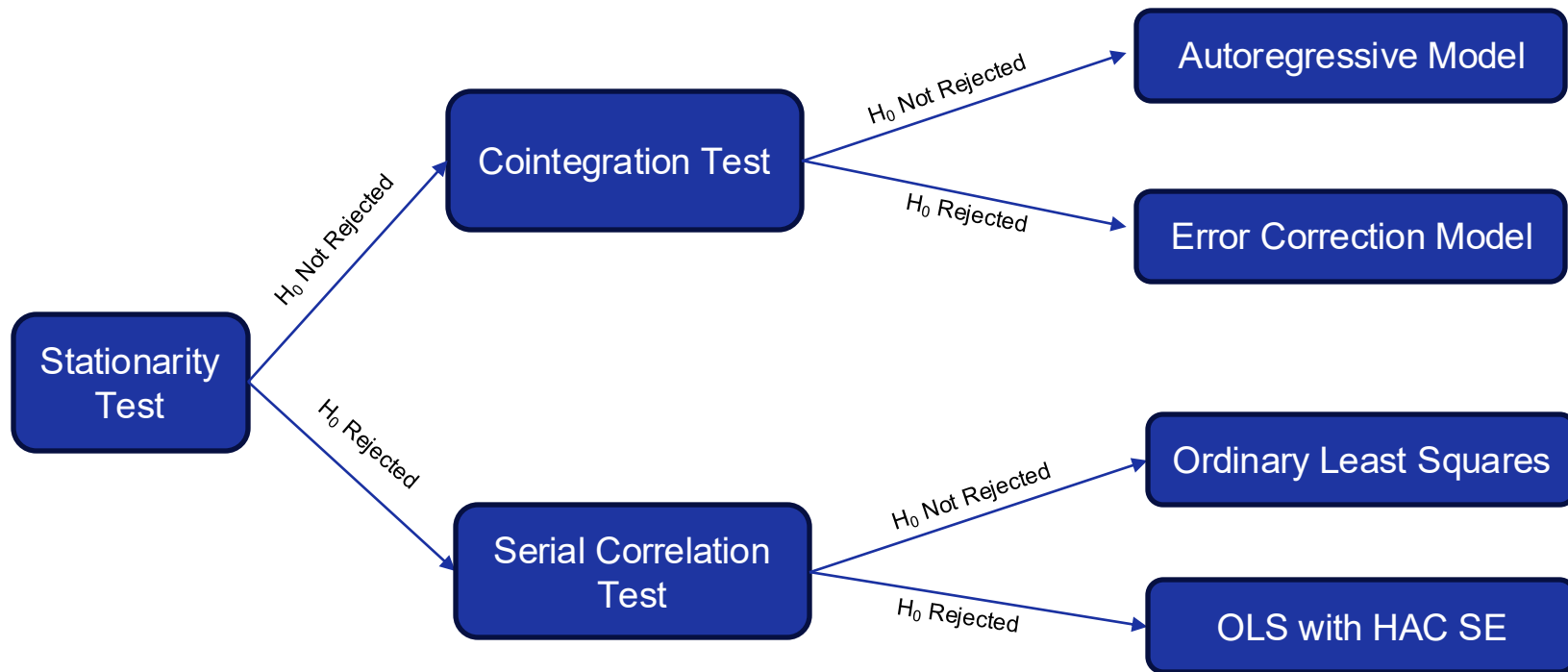
REM2 Research Methods

Lecture 7: Time Series Analysis

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A Road Map



Stationary and Nonstationary Variables

- The time series y_t is stationary if for all values, and every time period, it is true that:

$$E(y_t) = \mu \quad (\text{constant mean})$$

$$\text{var}(y_t) = \sigma^2 \quad (\text{constant variance})$$

$$\text{cov}(y_t, y_{t+s}) = \text{cov}(y_t, y_{t-s}) = \gamma_s \quad (\text{covariance depends on } s, \text{ not } t)$$

- Nonstationary series with nonconstant means are often described as not having the property of **mean reversion**
- Stationary series have the property of mean reversion

Stationary and Nonstationary Variables

- The econometric model generating a time-series variable y_t is called a **stochastic** or **random process**
- A sample of observed y_t values is called a particular **realization** of the stochastic process
- An AR(1) model

$$y_t = \rho y_{t-1} + v_t$$

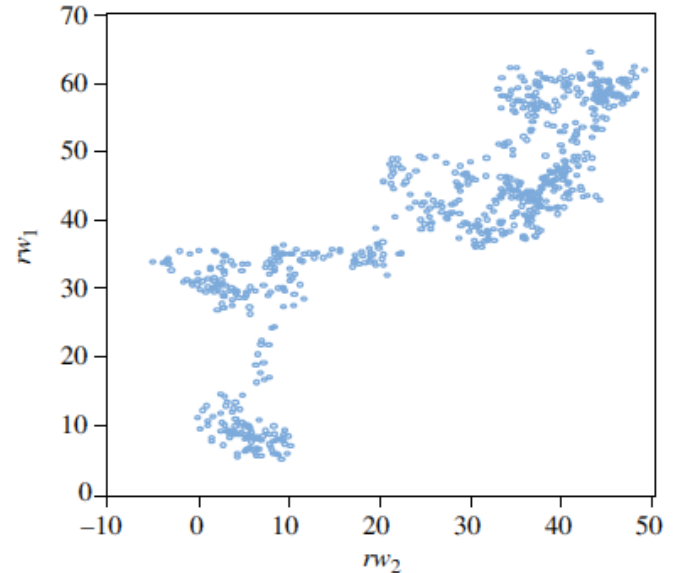
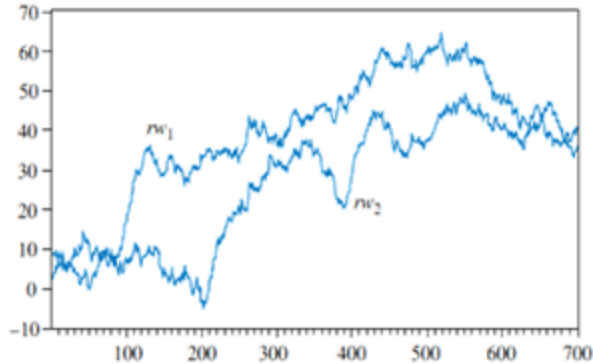
- The errors v_t are independent, with zero mean and constant variance σ_v^2 , and may be normally distributed
- The errors are sometimes known as “**shocks**” or “**innovations**”
- It is stationary if $|\rho| < 1$

Spurious Regressions

- Apparently significant regression results from unrelated data when nonstationary series are used in regression analysis
- Consider two independent random walks:

$$rw_1 : y_t = y_{t-1} + v_{1t}$$

$$rw_2 : x_t = x_{t-1} + v_{2t}$$



Spurious Regressions

- A simple regression of series one (rw_1) on series two (rw_2) yields:

$$\widehat{rw_{1t}} = 17.818 + 0.842 rw_{2t}, \quad R^2 = 0.70$$

(t) (40.837)

- These results are completely meaningless, or spurious. The apparent significance of the relationship is false
- The least squares estimator and least squares predictor do not have their usual properties, and t -statistics are not reliable

Unit Root Tests for Stationarity

- The AR(1) process $y_t = \rho y_{t-1} + v_t$ is stationary when $|\rho| < 1$
- We want to test whether ρ is equal to one (nonstationary) or significantly less than one (stationary)
- Tests for this purpose are known as **unit root tests for stationarity**
- Dickey–Fuller test is the most commonly used one
- The null hypothesis is non-stationarity.

Cointegration

- Definition: If $e_t = y_t - \beta_1 - \beta_2 x_t$ is a stationary $I(0)$ process, y_t and x_t are said to be **cointegrated**
- Cointegration implies that y_t and x_t share similar stochastic trends, and, since the difference e_t is stationary, they never diverge too far from each other
- In this case regression analysis is valid. No concerns about spurious regression.
- The test for stationarity of the residuals is based on the test equation:
$$\Delta \hat{e}_t = \gamma \hat{e}_{t-1} + v_t$$
- Critical values vary according to regression model

Error Correction Model

- Consider a regression model with nonstationary variables:

$$y_t = \delta + \theta_1 y_{t-1} + \delta_0 x_t + \delta_1 x_{t-1} + v_t$$

- If y and x are cointegrated, it means that there is a long-run relationship between them
- To derive this exact relationship, we set $y_t = y_{t-1} = y$, $x_t = x_{t-1} = x$ and $v_t = 0$, we get

$$y(1 - \theta_1) = \delta + (\delta_0 + \delta_1)x$$
- Re-arrange the ARDL model by adding the term $-y_{t-1}$ to both sides of the equation, and adding the term $-\delta_0 x_{t-1} + \delta_0 x_{t-1}$ to the RHS, we obtain

$$\Delta y_t = \delta + (\theta_1 - 1)y_{t-1} + \delta_0(x_t - x_{t-1}) + (\delta_0 + \delta_1)x_{t-1} + v_t$$

- Manipulating this we get $\Delta y_t = (\theta_1 - 1) \left(\frac{\delta}{(\theta_1 - 1)} + y_{t-1} + \frac{(\delta_0 + \delta_1)}{(\theta_1 - 1)} x_{t-1} \right) + \delta_0 \Delta x_t + v_t$, or ,

$$\Delta y_t = -\alpha(y_{t-1} - \beta_1 - \beta_2 x_{t-1}) + \delta_0 \Delta x_t + v_t$$

Regression When There Is No Cointegration

- We need to convert nonstationary, non-cointegrated series to stationary series
- The technique depends on whether the variables are
 - difference stationary ($y_t = \alpha + y_{t-1} + v_t$) or
 - trend stationary ($y_t = \alpha + \delta t + v_t$)
- Difference stationary: first differencing
$$\Delta y_t = \alpha + v_t$$
- Trend stationary: de-trending $y_t - \alpha - \delta t = v_t$

Stationary and Nonstationary Variables

- If the variables are $I(1)$ and not cointegrated, we need to estimate a relationship in first differences, with or without the constant term
- If they are trend stationary, we can either de-trend the series first and then perform regression analysis with the stationary (de-trended) variables or, alternatively, estimate a regression relationship that includes a trend variable
- If variables are stationary, or $I(1)$ and cointegrated, we can estimate a regression relationship between the levels of those variables without fear of encountering a spurious regression

Regression with stationary data

- When modeling relationships between variables, the nature of the data that have been collected has an important bearing on the appropriate choice of an econometric model
 - Two features of time-series data to consider:
 1. Time-series observations on a given economic unit, observed over a number of time periods, are likely to be **correlated**
 2. Time-series data have **a natural ordering** according to time

Regression with stationary data

- The primary assumption in OLS is:

$$\text{cov}(y_i, y_j) = \text{cov}(e_i, e_j) = 0 \quad \text{for } i \neq j$$

- For time series, this is written as:

$$\text{cov}(y_t, y_s) = \text{cov}(e_t, e_s) = 0 \quad \text{for } t \neq s$$

- The dynamic models mentioned previously imply correlation between y_t and y_{t-1} or e_t and e_{t-1} or both, so they clearly violate this assumption

Regression with stationary data

- Least squares estimation without recognizing the existence of serially correlated errors
 - The least squares estimator is still a linear **unbiased** estimator, but it is no longer best
 - The formulas for the **standard errors** usually computed for the least squares estimator are no longer correct
 - **Confidence intervals and hypothesis tests** that use these standard errors may be misleading

Serial Correlation

- When a variable exhibits correlation over time, we say it is **autocorrelated** or **serially correlated**
- How do we test whether an autocorrelation is significantly different from zero?
 - The k -th order sample autocorrelation
 - Correlogram (the sample autocorrelation function)
 - Lagrange Multiplier test
 - Durbin-Watson test (used less frequently today)

Serial Correlation

- The k -th order sample autocorrelation test
- The null hypothesis is $H_0: \rho_k = 0$
- The test statistic is: $Z = \frac{r_k - 0}{\sqrt{1/T}} = \sqrt{T} r_k \sim N(0,1)$
- The sample statistic, or the **k -th order sample autocorrelation** for a series y that gives the correlation between observations that are k periods apart, is:

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$

Serial Correlation

- LM Test: a joint test of correlations at more than one lag
- In a simple linear regression model

$$y_t = \beta_1 + \beta_2 x_t + e_t$$

- If residuals are correlated, then one way to model the relationship between them is to write:

$$e_t = \rho_1 e_{t-1} + \rho_2 e_{t-2} + \dots \rho_p e_{t-p} + v_t$$

- Test $H_0: \rho = 0$ by using a linear regression equation

$$y_t = \beta_1 + \beta_2 x_t + \rho_1 \widehat{e}_{t-1} + \rho_2 \widehat{e}_{t-2} + \dots \rho_p \widehat{e}_{t-p} + v_t$$

$$\widehat{e}_t = (\beta_1 - b_1) + (\beta_2 - b_2)x_t + \rho_1 \widehat{e}_{t-1} + \rho_2 \widehat{e}_{t-2} + \dots \rho_p \widehat{e}_{t-p} + v_t$$

$$T \square R^2 \sim \chi^2(p)$$

Regression with serial correlation

- Three estimation procedures are considered:
 1. Least Squares (LS) estimation with Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors
 2. Nonlinear Least Squares with an AR(1) error specification

$$e_t = \rho e_{t-1} + v_t$$

3. Autoregressive Distributed Lag (ARDL) model

Regression with serial correlation

- LS with HAC standard errors:

$$y_t = \beta_1 + \beta_2 x_t + e_t, \text{ where } e_t \text{ is serially correlated}$$

- HAC (heteroskedasticity and autocorrelation consistent) standard errors, or Newey-West standard errors

- Analogous to the Heteroskedasticity Consistent (HC) Standard Errors

$$\widehat{\text{var}}_{HAC}(b_2) = \widehat{\text{var}}_{HC}(b_2) \times \hat{g}$$

$$\hat{g} = \left[1 + \frac{\sum_{t \neq s} w_t w_s \text{cov}(e_t, e_s)}{\sum_t w_t^2 \text{var}(e_t)} \right]$$

$$\widehat{\text{var}}_{HC}(b_2) = \sum_t w_t^2 \text{var}(e_t)$$

$$w_t = (x_t - \bar{x}) / \sum_t (x_t - \bar{x})^2$$

Exercises

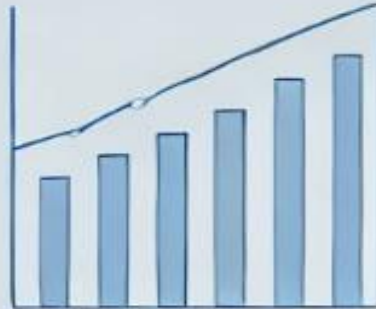
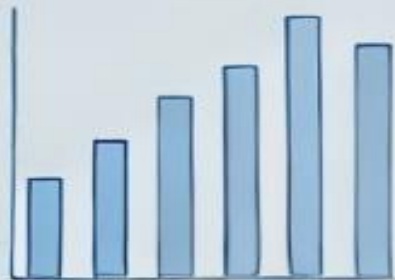
1. Open “Lectures 4-6 Exercises Data.xlsx” in a software of your choice. Use the data in the “Lecture 6” worksheet in this exercise.
2. Generate time series charts for the return and price series. Arrange the charts in a 2 by 3 grid.
3. Plot REIT_price, Anxious_index and CPI in one time series line chart, using quarter as the horizontal axis label.
4. Check the stationarity of each variable.
5. Estimate the following linear regression model:

$$REIT_return_t = \beta_0 + \beta_1 Anxious_index_t + \beta_2 Real_GDP_t + \varepsilon$$

6. Estimate the following autoregressive model

$$REIT_return_t = \beta_0 + REIT_return_{t-1} + \beta_2 Anxious_index_t + \beta_3 Real_GDP_t + \varepsilon$$

7. Check integration among REIT_price, Anxious_index, CPI Real, and_GDP.
8. Estimate a vector autoregressive model with REIT_price, Anxious_index, CPI Real, and_GDP.
9. Estimate a vector error correction model with REIT_price, Anxious_index, CPI Real, and_GDP.



Q&A

